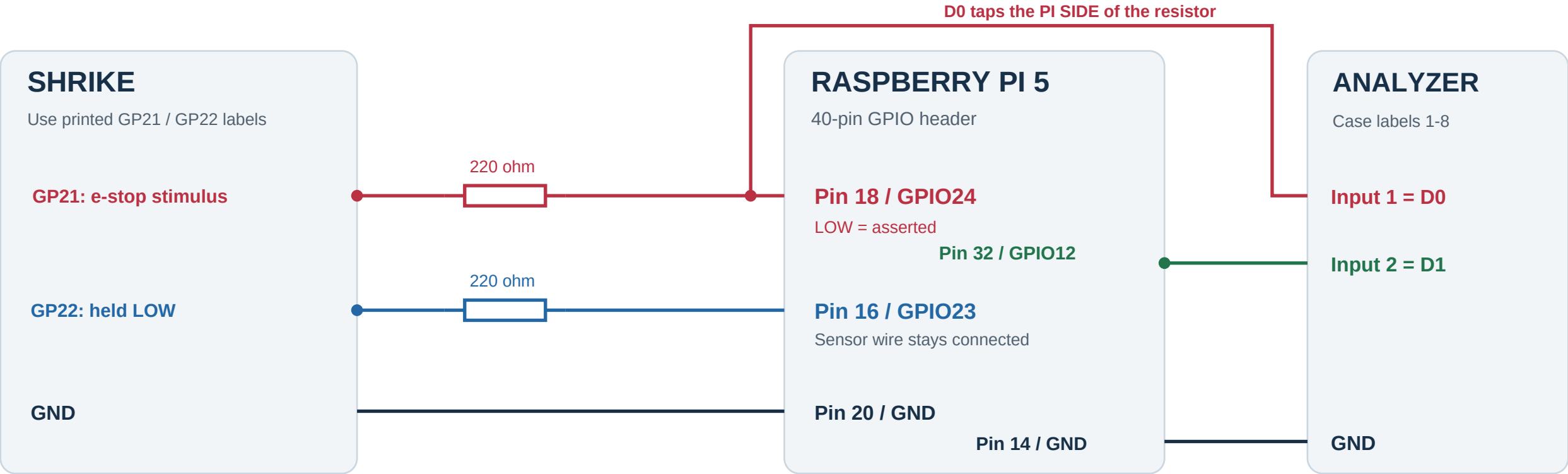


axiomos / physical e-stop test

Pi 5 + Shrike stimulus + logic analyzer + Debug Probe | No motors, motor drivers or actuators

WIRE WITH BOTH BOARDS UNPOWERED. All GPIO signals are 3.3 V. Do not connect a power rail to GPIO24.

Signal diagram - functional layout, not board orientation. Pi labels are PHYSICAL header pin numbers.



Only change from sensor testing: move analyzer input 1 (D0) from pin 16 to pin 18. Input 2 (D1) stays on pin 32.

Pi pins 14 and 20 share ground internally. Use the same resistor values and existing ground connections.

USB, UART, SD and power

Host USB -> Shrike USB

Host USB -> Logic analyzer USB

Host USB -> Debug Probe USB -> probe UART port (U) -> Pi dedicated 3-pin UART connector

Keep the existing keyed UART cable unchanged; do not move it to GPIO14/15 on the 40-pin header.

SD card -> Pi microSD slot

Pi USB-C power supply -> Pi power port (OFF until script prompt)

Power-on order

1. Pi power OFF + Shrike USB unplugged: insert the prepared SD and make the wiring above.
2. Connect Shrike USB, analyzer USB and Debug Probe USB. Leave Pi power OFF.
3. Run the e-stop capture script. Power the Pi ONLY at "UART RECORDING - POWER ON THE PI NOW".
4. Keep hands off during automatic presses. Power off Pi at completion. Do not attach actuators.

Test image: bench-estop-rearm. Automatic output rearm on release is diagnostic-only; this is not the FPGA/motor cutoff test.

Source: campaign wiring + scripts/hil/v03d-estop-cycle.sh; Pi pin numbering / debug header: official Raspberry Pi documentation.